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museums

Editor: Halina Gottlieb
The Interactive Institute

Halina Gottlieb
David Nilsson

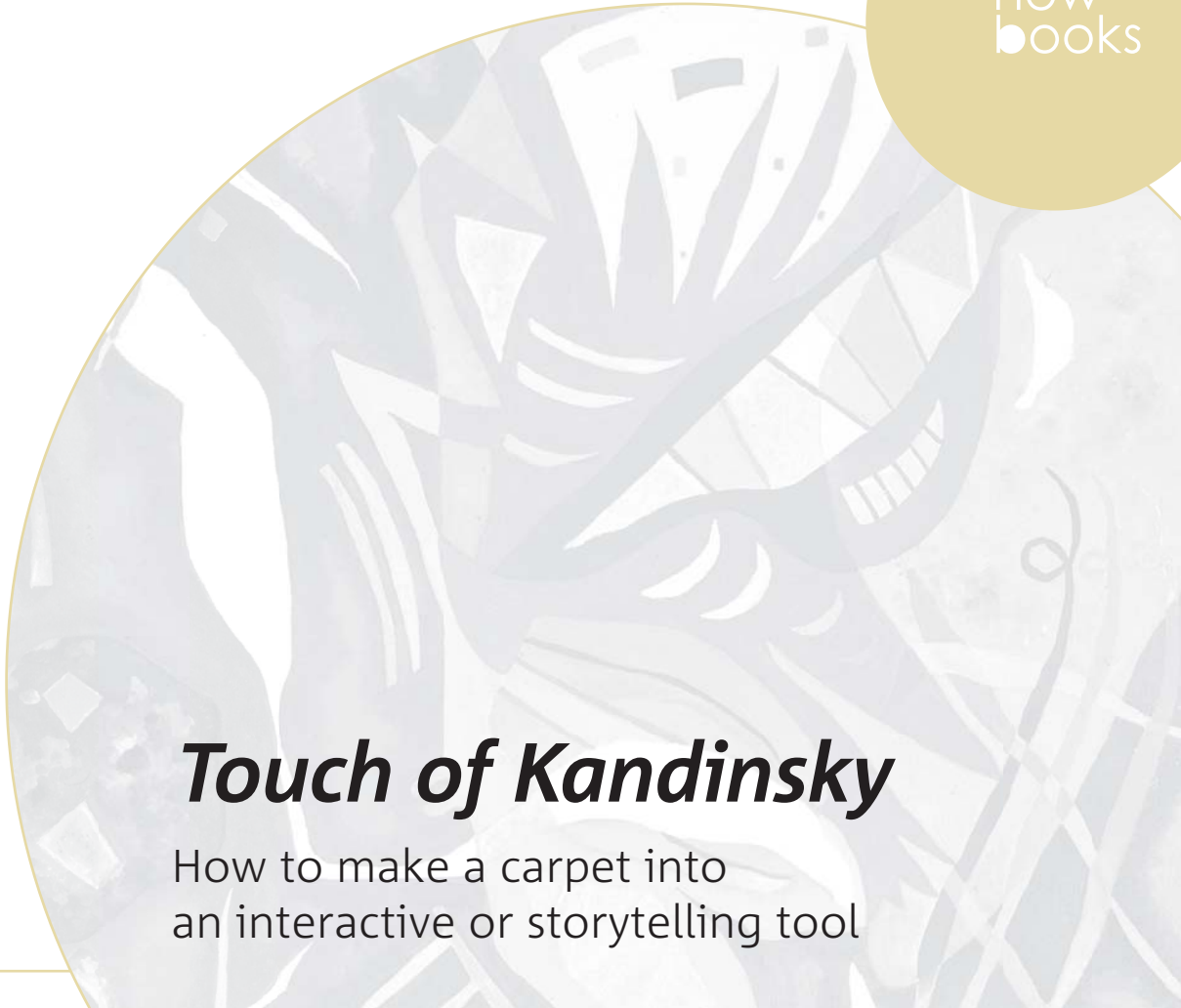
Touch of Kandinsky

How to make a carpet into an interactive or storytelling tool



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Touch of Kandinsky

Touch of Kandinsky is an installation in the form of an interactive carpet, aiming to increase the understanding of abstract art. The carpet is primarily intended for small children visiting art museums or galleries. It is inspired by the artist Wassily Kandinsky's writings on the concept of synaesthesia and is based on his painting *Group* from 1937. The carpet contains built-in sensors that create an interactive field. The interactivity is primarily limited to the figures in the motif and the sound they create. The sensors under the thick-woven surface activate various sounds when the user touches, walks, presses or rolls over the carpet. Depending on the pressure and the length of time it is applied, the sound subtly twirls and changes.

The carpet is an experiment in communication, exploring the personal dimensions of the artist's theory through interactive discovery. The project is conducted at the Interactive Institute and EPOCH.

This KNOWHOW booklet will also give you a short description of the implementation of

a project called the *Whispering Carpet* based on the same technical solution as *Touch of Kandinsky,* but with a different approach and implementation method. (See the chapter *Implementation: Whispering carpet*)

The visitor's experience

Generally the main focus at traditional museums is a presentation of the museum's collections of objects and of the employees' knowledge of these collections. However, as a consequence of development, interest has shifted during the last years from a focus on the museum itself to the stimulation of visitors' curiosity and the encouragement of interactive learning where they can engage in their own experiences and discoveries. It is now often desired that visitors themselves recognize the multiplicity of their needs and interests.

Museums' initial interest in digital media was essentially about the technology. To a lesser extent it was about how digital media tools can be utilised as multimodal tools alongside the more traditional options in the form of signs, exhibits, catalogues and exhibition stands. Now there is an effort to combine these two aspects: visitor orientated exhibitions and digital media that can engage visitors at the museum.

An interactive tool in the form of a carpet with the motif of a painting can be a playful complement to the art galleries, which often challenge visitors' perceptions. By encouraging visitors to interact with the motif on the carpet, the museum can enrich the experience of abstract art for both grown-ups and children alike. The carpet can be an important addition to the arranged curator-led activities in the gallery, offering an area for self-directed exploration. Attracting the visitor's attention to the other paintings, the carpet can stimulate communication between visitors, both young and old, sharing the experience of their simultaneous exploration.

Breaking it down: the components of *Touch of Kandinsky*

With the exception of the actual carpet, *Touch of Kandinsky* consists of several components that make up the installation. When exhibiting the installation the carpet is placed on a podium consisting of 28 boxes, each sized W53 x D53 x H22 cm. Once put together they form a stage-like podium upon which the carpet is placed. There are six sets of sensors placed under six motifs on the carpet, under which six midi-speakers are placed. The installation runs on a PC with a midi-interface card and external sound card connected. All of the equipment is placed underneath the podium. The installation can be turned on/off through a hatch in one of the edge boxes housing most of the technical components. If this edge box is facing a wall, it could be left open to increase the airflow under the podium. When transported, the boxes along with the electronic equipment are placed in 2 large crates. The carpet with the attached sensors is placed in a rectangular box on wheels for easy transportation.

How it works: Experience *Touch of Kandinsky*

If, for example, a visitor walks on the mosquito figure, a short circuit is created when the aluminium foil-coated papers (the sensors) connect. A signal is sent to the MIDI-interface card and the card transforms the short circuit into a MIDI-signal. The signal is then transferred to an external soundcard and through a firewire cable to the computer. On the computer the tool Ableton LIVE recognises the signal as the mosquito signal through the MIDI-protocol. The mosquito track is activated and sends the sound to the external soundcard. The card distributes the sound of the mosquito through the speaker underneath the mosquito figure.





Figure ► Sensor ► MIDI-interface card ► External soundcard ► Computer ► External Soundcard ► Speaker ► Figure





Images from the production of the carpet at Hitex AB's factory in Svängsta, Sweden.

Manufacturing the carpet

In order to transfer the abstract painting on to a large carpet we contracted the Swedish-based company Hitex AB. They have great experience in the manufacture of custom-made carpets and have developed their own robotic system for creating quality hand tufted carpets.

The Sensors

When developing the installation, we selected six of the painting's motifs that we wanted to make interactive. Since the painting has several abstract figures we wanted to create custom-sized sensors that only respond to the user when one of the six chosen figures is touched or walked on. We came up with an in-house solution that worked perfectly and also made the sensors cheap and easy to build.

The sensors consist of two layers of aluminium foil-coated paper with perforated plastic foam in between. When pressure is added, the foil surfaces touch through the perforated foam and thus work as a switch.

Building the sensors

Materials and tools used for creating the sensors:
– Aluminium foil-coated paper
– Plastic foam (2-4 mm thick)
– Wire 2-core (thin)
– Soldering iron
– Solder
– Printer
– Computer
– Photoshop (or similar design tool)
– Scissors
– Cutting knife
– Glue gun with glue sticks
– Duct tape



The thick paper we used for the sensors is coated on one side with aluminium foil. This particular paper is originally use by plumbers for insulating pipes, but any similar paper with aluminium foil can be used. The important thing is to ensure that



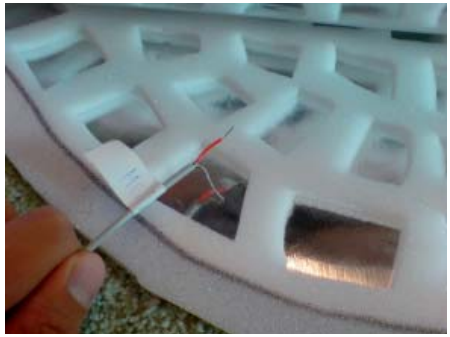
the aluminium foil can endure frequent use without easily being worn out.
The first step in creating the sensors is to print the motif in 1:1 size and use the print as a template to cut out the figures from the aluminium foil-coated paper. Two copies of this paper need to be placed with the aluminium sides facing each other.

When the carpet was produced, a digital version of the carpet was drawn up in **CAD** (a design tool, short for Computer-Aided Design) and uploaded to the robotic system. Hitex AB provided us with this digital sketch in a PDF format. You will need to slice the PDF document of the painting into A4 or A3 squares in Adobe Photoshop (or any other design tool you are familiar with) and print the carpet in 1:1 size. Another option would be to find a professional print shop to do a large format print, but this will probably

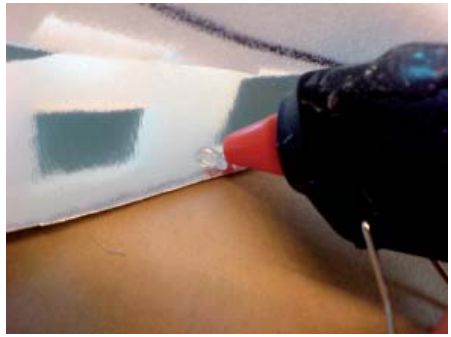












be more expensive than printing the image internally.

After cutting out the aluminium foil-coated paper, you cut out a slightly larger version (20-30mm) of the figures in plastic foam. In order to get the aluminium papers to touch each other when pressure is added, you need to make small holes in the plastic foam. The holes should be about 30x50mm in irregular forms approximately 20mm apart. The plastic foam should not be too thick, 2-4mm is recommended. We used 4mm thick plastic foam that works perfectly with the size of the holes mentioned above.

The important thing is that the aluminium papers connect if pressure is added. Therefore the size of the holes, thickness of the plastic foam and the weight of the carpet are all parameters to take into account when creating the sensors. If the holes are too small it will be difficult for the aluminium papers to make contact. If the holes are too big you will run the risk of the aluminium papers remaining connected even after the pressure is removed from the sensors. Make sure to leave a good distance/border (30-40mm) from the perforated area to the edge of the plastic foam. The aluminium paper should be larger than the perforated area, but smaller than the entire foam cutout. A tip would be to make a test sensor and try it out before you start making them all.

Before gluing the sensors together, you will need to cut the wires that are going to connect the sensors to the computer through the MIDI and the sound card. Make sure the wires are not too short to reach the MIDI-interface card, wherever you plan to place it. It is best to cut the wires a bit longer. Use a wire that is as thin as possible with 2 cores. Take one end of the wire and strip off about 80mm of its outer plastic cover. Strip the two conductors from their plastic protective covering so that about 30-40mm

is exposed. Attach one conductor to each aluminium paper. We used duct tape to attach the conductors. Glue the aluminium papers to the plastic foam, one at a time (the duct tape didn't show to be very reliable for this assignment).

When the wires are properly attached and the sensor is working, it is time to attach the sensor to the carpet. In order for the sound to travel through the sensors and the carpet with as little deterioration as possible, we made the carpet thinner at the points of contact with the sensors. By doing this we created areas of least resistance for sound to travel through.

Now that the sensors are made and tested, they can be mounted underneath the carpet. Since the underside of our carpet has a plastic surface we chose to glue the sensors to the back of the carpet using a glue gun. To protect the sensors and the wires we dressed the back of the carpet in thick plastic. Make sure to cut holes in the plastic where the speakers are placed to avoid suffocating the sound.

Connecting the sensors

We suggest that someone with experience in soldering does this part of the job because it is too complex to describe in this KNOWHOW booklet. In preparation for soldering, the wires from the sensors are placed in a desirable order and attached to the carpet with duct tape. Make sure the ends of the wires that are going to be connected to the MIDI-interface card meet at the point you decide to place the card. Even though the distance from the carpet to the box containing the MIDI-interface card is no more than a couple of decimetres away, we recommend that you make sure the wires coming out from underneath the carpet, leading to the point of contact are at least 1.5m long. This makes it easy to handle the carpet when setting it up or supporting it.

We used a standard contact equipped with 10 solder connection pins. You will need to use seven of them, six for one conductor from each wire. On the seventh pin you solder the second conductor from each of the six wires, creating the earth.

From the MIDI-interface card you attach a flat cable to pin 1-6 (supports up to 48 altogether). The other end of the flat cable you solder to the contact. Strip six of the wires on the flat cable and solder these wires to six different pin connections on the card. Strip one of the wires from the flat cable and solder it to the earth connection on the MIDI-interface card. Now each wire from the sensors should be connected to both earth and to a single pin connection (channel) on the midi-card.

Composing the sound

A musician composed the sounds with inspiration from Kandinsky's writings on synaesthesia. In *Concerning the Spiritual in Art* Kandinsky writes: "The light red awakens a feeling of power, joy, determination, of a loud triumph, etc. The color reminds one, musically, of a fanfare of tubas in the lowest part – a strong, stubborn and obstinate sound."

The musician and selected children were invited to a workshop to associate the shapes and colors of the carpet with possible sounds they could make. Eight sounds were associated with an equal amount of figures.

These were recorded and we chose the six sounds that we felt worked the best. When placing six speakers in a relatively small area (such as the carpet), it is necessary to place them so that they are not too close to each other, which could lead to the sounds clashing. When composing the sounds it is important to keep in mind that the different sounds should interplay with each other in the best possible way when activated simultaneously.

MIDI-interface and soundcard

MIDI (Musical Instrument Digital Interface) is an industry-standard protocol that enables electronic musical instruments, computers and other equipment to communicate, control and synchronise with each other. MIDI does not transmit an audio signal or media, it simply transmits digital data 'event messages' such as the pitch and intensity of musical notes to play and control signals for parameters such as volume and panning.

We used a MIDI-interface card called **CTM64**, which is a card used to generate MIDI note on/off or program change messages. It is suitable to retrofit keyboards, switch panels, button arrangements or any other contacts so that they are able to transmit MIDI data. The CTM64 cannot be used independently but has to be combined with additional electronic equipment to become a working device (e.g. keyboard, switches, buttons, potentiometers, power supply). When purchased, the card comes without a hard-case box, so for the protection of the card we recommend you purchase such a box.

The soundcard we chose is called **M-AUDIO Firewire 410** and has 8 output connections and is connected to the PC via a firewire cable. It also has in and out MIDI connections that enabled us to connect the MIDI-interface card.

Programming the sound in Ableton LIVE

When the sound is composed and recorded you need to import the sound files in a music sequencer tool. We used **Ableton LIVE** for this task after considering price and ease-of-use factors. LIVE is a loop-based software music sequencer for both Mac and Windows and is designed around the concept of being as much an instrument for live performances as a tool for composing and arranging.
The sound files were imported to LIVE and six different tracks/channels were created, one for each sound/sensor. Each track has its own output channel and speaker connected to this channel via the soundcard. We programmed the application to play a specific sound clip when receiving a MIDI signal from one of the sensors. Each sensor is tagged with a unique midi signal through the



Screenshot of Ableton LIVE

MIDI-interface card, and when LIVE receives the signal it plays the sound-clip connected to the MIDI signal. To avoid clean cuts and sudden starts of the sounds we added a 1-2 second fade in and fade out function. To create the fade function you add a sampler called 'simpler' (included in LIVE) to the tracks. If you want customised fades for each sound you need to create a MIDI-filter for each track.

PC and OS set-up

For this installation we used a modern medium sized PC with an external 15" screen. The screen is hidden inside one of the boxes forming the podium that the carpet is lying on and is only used for set-up and troubleshooting. A laptop with a firewire connection would also work, but since the screen is only used during set-up and troubleshooting a small desktop computer fits the purpose better in this case. A good choice of computer could be the **Mac Mini**, which is small, has proper connections, is quiet and stays relatively cool.

The computer we used runs Windows XP and was prepared for 'kiosk mode':

- All desktop animation features disabled, OS set to 'best performance'
- All energy saving settings off
- Standard OS services that enable redundant tasks such as surfing the web disabled
- Software updates and error reporting disabled
- Login set to 'no password' mode
- Booting of computers set to minimal features, no graphic splash screens

Implementation: The Whispering Carpet

With inspiration from the *Touch of Kandinsky* project, a new museum installation at the Archaeological Museum in Stavanger was developed in 2006 called the *Whispering Carpet*.

Using a modern copy of an antique Persian carpet, the Swedish Forum for Cultural Heritage at the Interactive Institute, in collaboration with Archaeological Museum in Stavanger, has added interactivity and created an experience room around it. The carpet





is triggered by the visitor standing, sitting or lying on the carpet, and tells the legend of Rustam in a voice from within the carpet. The legend is a tragic drama about the great warrior Rustam who ends up killing his own son in battle. The characters and elements from the story are depicted in the carpet's motif, and by forming optical light threads around the figures' contours it emphasizes those elements when appearing in the story. The optical light and three programmed LED-light walls follow the mood of the story as it evolves from romance to drama.

When entering the room, the visitors are faced with a fairly dark environment with subtly changing LED-light walls and atmospheric sounds. The carpet is placed on a 40cm high platform at one end of the room. Visitors can reach the carpet via the stairs, or a ramp placed in connection to the platform for easy accessibility. The carpet is partly shielded by a transparent fabric hanging from the roof. This fabric creates a 'room within the room' and provides a cosy atmosphere for the visitors entering the 'room'. When a visitor walks onto the carpet, it activates and the story begins. If the visitor steps away from the carpet, the story will pause. If the carpet is reactivated within 10 seconds, the story will be continued from were it was paused. If not, the story starts from the beginning next time the carpet is activated.

The sound of the installation can be broken down in two parts. The main sound is the narrative voice telling the story which comes from underneath the carpet. The other sounds, which fill the room and follow the story, are atmospheric sounds and sound effects. These sounds are distributed through four speakers placed behind the LED-light walls and convey ambient noise from the story's environment, such as sounds of the forest, the battlefield, a running horse etc.

Working with fibre optic light threads

Fibre optic light threads are plastic fibres designed to guide light along its length and to let it out at the end of the thread. But if you rub the surface of the fibre you will let the light out before it reaches the end of the thread, creating a glowing thread.

There are many different sizes and capacities of fibre optic light threads. It is important to test the varying options for suitable use, depending on the carpet, need of light, flexibility etc. We used a thread that is 1.5mm thick with fairly elastic qualities. The thread can be purchased in a bunch of hundreds of threads in desirable lengths connected to a contact that you plug into a strong halogen lamp, feeding the threads with light and making them glow. A switch on the lamp is then connected to a similar MIDI-interface card used in the *Touch of*





Kandinsky installation and connected to the computer that is programmed to turn the lamp on and off on specific occasions.

By experimenting with fibre optic light threads, we wanted to emphasize the characters depicted in the carpets motif when appearing in the story. To do this we attached the threads to the figures' contours and sewed about 20-30cm of the fibre into the carpet leaving the rest of the thread and the contact underneath the carpet.

In order to sew the thread into the carpet, you need to place it in the contours of the figure whilst making sure it doesn't disappear into the carpet and lose visibility. Use a strong needle and fishing thread to attach the optics to the carpet and then carefully rub the fibre optic light thread with a fine-grained sandpaper. More light will be released at the points you rub the hardest, and therefore less light is left for the rest of the optical thread. If you rub the thread before you attach it to the carpet, you will have to rub around the circumference of the thread. If you attach it first, only the surface that is visible needs to be rubbed, leaving more light for the remaining length of the thread.

Visitor studies

Visitor studies have been carried out on several occasions. The aim of these studies was to observe children's ability to experience an abstract painting presented in this manner. This was the main underlying question when planning observations and documentation of visitors' interaction with the *Touch of Kandinsky* installation.

The formative study showed that some visitors were positively surprised that the carpet actually had sound as an invisible dimension, while others found this perplexing, having difficulties in marrying their tactile experience with the sound effects. Many respondents compared the carpet to a musical instrument, ´but you can't see the strings´. The fact that the carpet was raised on a low platform gave it a stage-like appearance. Family members, usually parents observing their child, watched one another perform on the carpet.

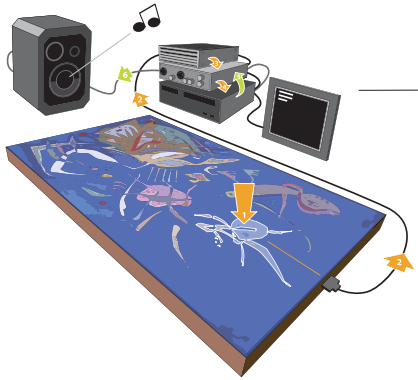
Unfortunately, the stage-like appearance may have discouraged or inhibited the more shy visitors from exploring the carpet. It also became clear that some visitors regarded the carpet as a piece of art that they weren't allowed to touch.

The interactivity of the carpet encouraged interaction and communication between the visitors. Visitors were exchanging smiles, chuckles and nods to each other while exploring the carpet. Children, in particular, often challenged one another by trying to cover as many active sound areas as possible with their limbs, or by adding their own songs in loud voices or by making funny noises.

Touch of Kandinsky

Technology/Interaction

In-house sensors connected to a PC respond to user interaction with the carpet's motif through a midi-interface card. Sounds linked to the motif are played out on external speakers placed under the carpet.



Ideas for Implementation

Used in connection with permanent exhibitions, entrance halls, coffee shops, walking halls, a children's atelier with the aim to:

- Stimulate curiosity, imagination and creativity about paintings and themes.
- Narrate history/stories related to motifs of art.
- Bridge the gap between an artist's message and visitor references.

Required competencies

Art Historian
Museums educator
Carpenter
Musician/Composer
Sound Designer
Interaction Designer
Stage Designer
Project Manager
Technician

Summary

Touch of Kandinsky is an installation in the form of an interactive carpet, aiming to increase the understanding of abstract art, primarily intended for small children visiting art museums or galleries.

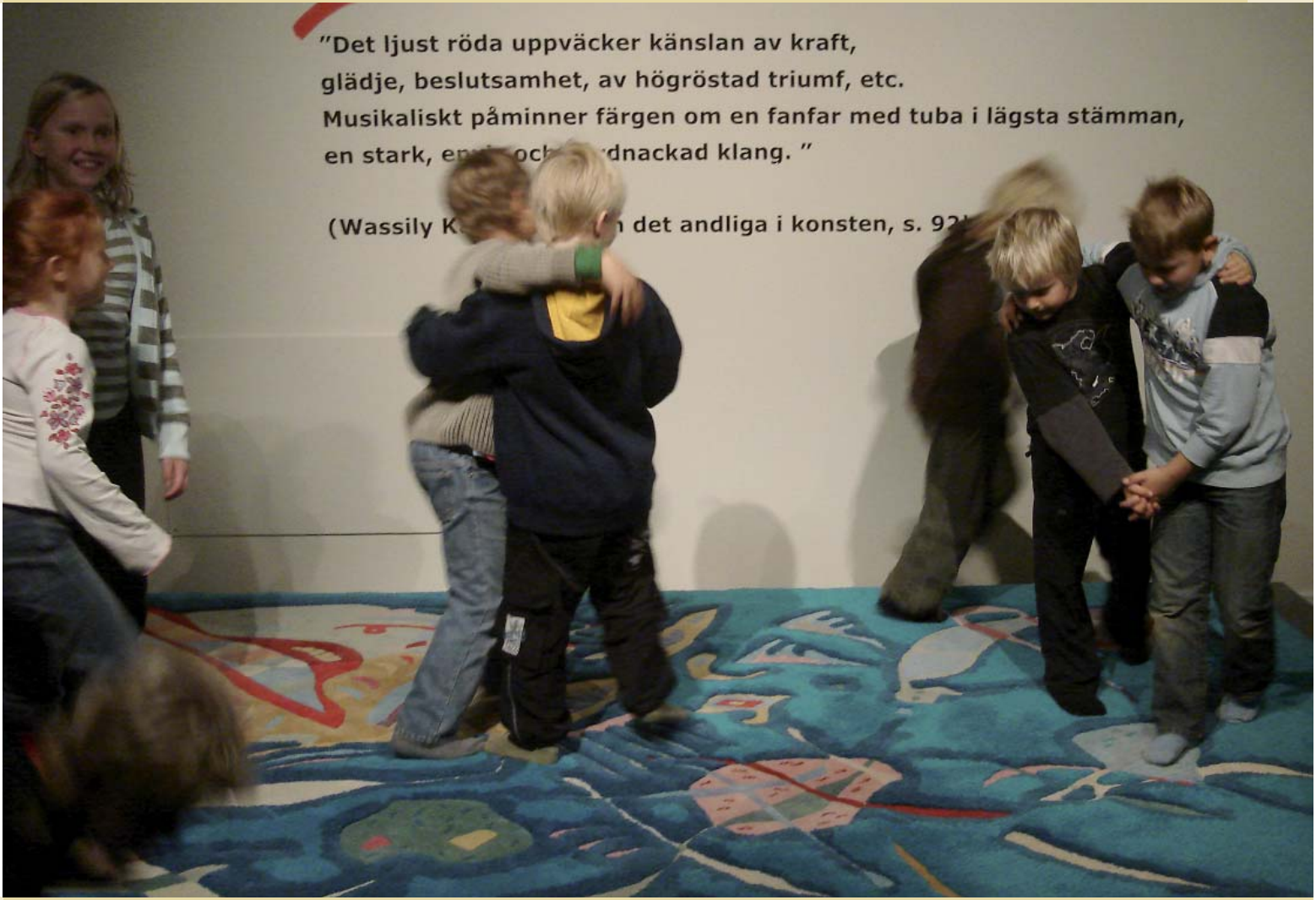
Resources

Person hours: 200

Technology: €2,500

Material: €200

Production costs for the carpet: €2,500, for the podium boxes €5,000


”Det ljust röda uppväcker känslan av kraft,
glädje, beslutsamhet, av högröstad triumf, etc.
Musikaliskt påminner färgen om en fanfar med tuba i lägsta stämman,
en stark,
dnackad klang. ”
det andliga i konsten, s. 92



Touch of Kandinsky

Project team

Initiator: Halina Gottlieb
Concept: Halina Gottlieb, Anna Enquist, Maria Taube
Project manager: David Nilsson
Music: Homero Alvares
Interaction Design: Bjarne Nykvist
Sound design: Stefan Lindberg
Technician: Alex Bustamante, David Jonsson
Stage Design: Anna Söderberg
Production of the carpet: Hitex AB
Illustrations for booklet: Andreas Beijer
Proof-reading for booklet: Kristi Wilson Lindroth

The Whispering Carpet

Project team

Initiator: Bente Ruud
Concept: Halina Gottlieb, David Nilsson, Lotte Hougaard, Bente Ruud
Project leader: Lotte Hougaard
Sound designer: Erland Eine
Light designer: Gunnar Aasland
Production: David Nilsson, Anna Söderberg, Lotte Hougaard
Interaction Design: Alex Bustamante
Dramaturge: Gabriella Thinsz
Museum educationalist: Siv Kristoffersson, Bente Ruud

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By encouraging visitors to interact with the motif on the carpet, the museum can enrich the experience of abstract art for both grown-ups and children alike. The carpet can be an important addition to the arranged curator-led activities in the gallery, offering an area for self-directed exploration. Attracting the visitor's attention to the other paintings, the carpet can stimulate communication between visitors, both young and old, sharing the experience of their simultaneous exploration.

The KNOWHOW booklets are an inspirational series cataloguing existing examples of a variety of projects which use ICT for the recording, display and interpretation of cultural heritage. These booklets highlight functional information covering the design, development and implementation of ideas and their solutions, and give thoughtful suggestions for alternative applications within the cultural heritage sector. The KNOWHOW booklets aim to support people working in the area of museums, heritage sites and monuments. The information covered within the booklets benefits managers, exhibition producers/curators, pedagogues and professionals working with digital restoration, as well as those working with communication and audiences. These booklets cover projects developed by the partners of EPOCH, and are divided into the following categories: MUSEUMS, HERITAGE SITES and MONUMENTS.

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